

Vascular Occlusion Emergency Protocol

Recognition, immediate response, hyaluronidase dosing, and post-event documentation

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VASCULAR OCCLUSION – SIGNS TO RECOGNIZE IMMEDIATELY

Sign	Timing	Description	Action
Immediate blanching	During injection	Sudden white discoloration – loss of blood flow to zone	STOP INJECTING NOW – this is the most critical sign
Severe out-of-proportion pain	During / immediately after	Burning, sharp, throbbing pain beyond expected injection discomfort	Stop. Reassess. High index of suspicion for occlusion
Dusky discoloration	Minutes after	Grayish-purple replacing blanching – progressing ischemia	URGENT – treat immediately with hyaluronidase
Livedo reticularis	Minutes after	Mottled net-like pattern following vascular territories	Strong indicator of vascular event – treat now
Skin temperature drop	Minutes after	Affected area noticeably colder than surrounding tissue	Supports diagnosis – begin treatment protocol
Vision changes – ANY	During / after	Blurring, darkening, shadow, diplopia	CALL 911 IMMEDIATELY – retinal artery occlusion

THE 7-STEP PROTOCOL – EXECUTE IMMEDIATELY

STEP 1 – STOP	Put the syringe down immediately. Do not inject another drop. Assess the patient and skin now.
STEP 2 – HYALURONIDASE	Inject 150–300 units (Hylanex preferred) throughout affected zone AND along vessel course. Do not wait. Have it in the room – always.
STEP 3 – MASSAGE + HEAT	Vigorous massage over affected area. Apply warm compress immediately to promote vasodilation. Continue throughout.
STEP 4 – ASPIRIN	Chew (not swallow whole) 325mg aspirin immediately if no contraindication. Antiplatelet effect reduces clot extension.
STEP 5 – CHECK VISION	"Is your vision normal in both eyes?" Any visual change = CALL 911 NOW. Retinal artery occlusion is irreversible within 90 minutes.
STEP 6 – REPEAT HYAL q60 min	If blanching/dyscoloration not improving at 60 min – repeat hyaluronidase 150u. Follow up every 24 hrs until resolved.
STEP 7 – CALL CARRIER	Contact your malpractice carrier as soon as patient is stabilized. Document everything – product, lot, volume, points, timeline.

HYALURONIDASE DOSING REFERENCE

Scenario	Dose	Product	Notes
Vascular occlusion — initial	150–300u	Hylanex (preferred) or compounded	Inject throughout affected zone AND along vessel course immediately
Vascular occlusion — repeat	150u every 60 min PRN	Same	Repeat until clinical improvement seen
Filler dissolution (elective)	30–150u per mL of filler	Per preference	Allow 2 wks for full resolution before assessing
Tear trough dissolution	15–30u per side	Dilute — use low concentration	Very conservative — less is more in tear trough
Nodule dissolution	10–30u per nodule	Per clinical judgment	Aspirate first if possible to confirm product

HIGHEST RISK ZONES — KNOW BEFORE YOU INJECT

Zone	Primary Risk Vessel	Blindness Risk	Preferred Technique
Glabella	Supratrochlear + supraorbital → ophthalmic	HIGHEST	Botox preferred. Filler: advanced only, tiny volumes, slow
Nasal dorsum/tip	Dorsal nasal + angular → ophthalmic	EXTREME	Advanced injectors only. Cannula. Very small volumes
Superior NLF / alar base	Facial artery (superficial here)	High	Cannula strongly preferred. Retrograde threading
Temple	Superficial temporal artery	Moderate	Palpate pulse. Cannula medially. Supraperiosteal deep
Lips — body	Superior/inferior labial arteries	Low-Moderate	Cannula from lateral. Serial puncture. Small aliquots
Tear trough	Infraorbital artery	Moderate	Cannula. Supraperiosteal medial only. Tiny volumes

POST-EVENT DOCUMENTATION — DOCUMENT ALL OF THIS

-  **PRODUCT:** Exact product name, lot number, expiration date, volume injected, injection points, depths
-  **TIMELINE:** Time of injection, first symptom, each intervention, each reassessment — minute by minute
-  **PHOTOS:** At event, each treatment interval, 24-hr follow-up, resolution or referral
-  **TREATMENT:** Hyaluronidase doses and timing, aspirin, warm compress, massage, any other interventions
-  **COMMUNICATION:** What you told patient, instructions given, referrals made, emergency services contacted
-  **CARRIER:** When you contacted malpractice carrier and what was communicated